

Comparing Sub-Populations

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→ Covered in Tutorial 9

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↑ Not covered. Will not be tested.

4.2.0 Preamble and Intuition

Overview

- We discuss the intuition behind comparing sub-populations
 - we use the shark encounter data as a motivating context.
-

Oftentimes, interest lies in comparing two or more sub-populations (i.e., $\mathcal{P}_1$ vs. $\mathcal{P}_2$).

- e.g., treatment vs. control arms of a clinical trial
- e.g., two different versions of a webpage in online A/B testing
- e.g., the Shark encounters that occurred in Australian vs. US waters

Suppose we have the population $\mathcal{P} = \{\mathcal{P}_1, \mathcal{P}_2\}$ and interest lies in comparing some attribute across the two sub-populations:

$$a(\mathcal{P}_1) \text{ vs. } a(\mathcal{P}_2)$$

- We could compare the two sub-population attributes by way of a difference:

$$a(\mathcal{P}_1) - a(\mathcal{P}_2)$$

- We could compare the two sub-population attributes by way of a ratio:

$$\frac{a(\mathcal{P}_1)}{a(\mathcal{P}_2)}$$

- If the attribute is graphical (a histogram or quantile plot, for example) we could compare the two sub-populations by displaying the figures beside one another or overlaying them on top of one another.

***Note:** that when the attribute is a **measure of location**, sub-population comparisons are typically based on differences and when the attribute is a **measure of spread**, comparisons are typically based on ratios.

Motivating Example: Comparing Australian vs. American Shark Encounters

- Let's load the data and numerically compare the sharks lengths in the two populations:

```
sharks <- read.csv("../Data/sharks.csv",
                  header = TRUE)
```

```
pop <- list(pop1 = subset(sharks, sharks$Australia == 1),
            pop2 = subset(sharks, sharks$USA == 1))
```

```
summary(pop[[1]]$Length)
```

Min. 1st Qu. Median Mean 3rd Qu. Max.
36.0 119.5 164.0 155.9 193.0 240.0

```
summary(pop[[2]]$Length)
```

Min. 1st Qu. Median Mean 3rd Qu. Max.
68.0 109.0 156.0 150.4 186.0 216.0

- Let's also graphically compare the shark lengths in the two populations using a quantile plot:

```
par(mfrow=c(1,3),oma=c(0,0,2,0))
```

```
qvals <- sort(pop[[1]]$Length)
pvals <- ppoints(length(qvals))
plot(pvals, qvals, pch = 19, col=adjustcolor("black", alpha = 0.5),
     xlim=c(0,1), ylim=extendrange(range(sharks$Length)),
     xlab = "Proportion p",
     ylab = bquote("Quantiles Q"["y"]*"("p)"),
     main = "Australia Encounters")
```

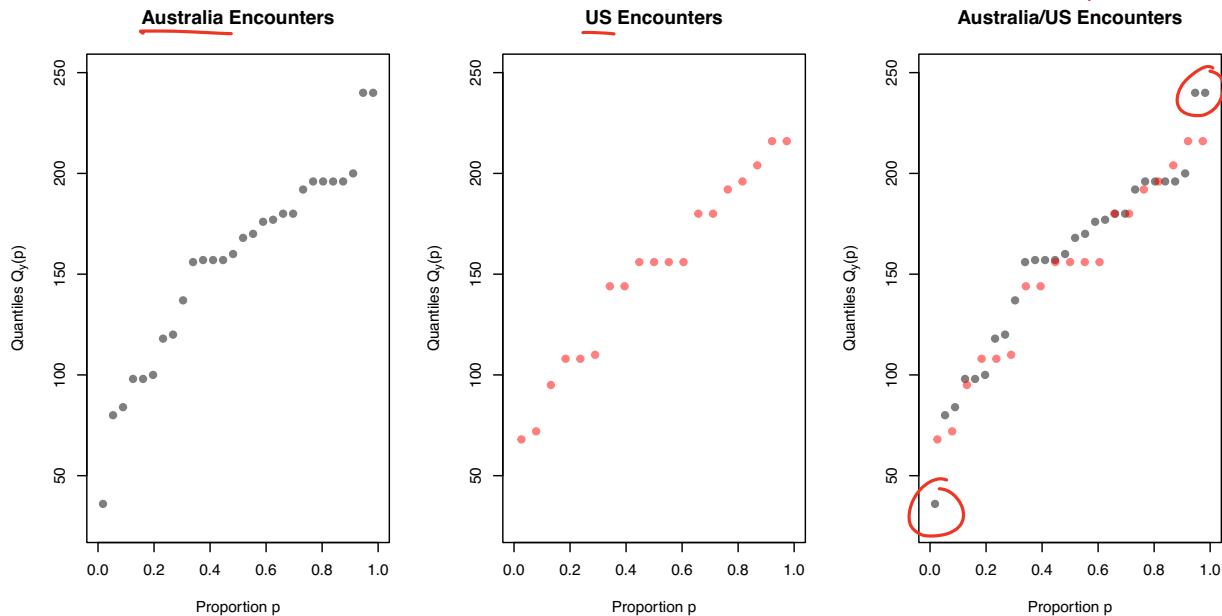
```
qvals <- sort(pop[[2]]$Length)
pvals <- ppoints(length(qvals))
plot(pvals, qvals, pch = 19, col=adjustcolor("red", alpha = 0.5),
     xlim=c(0,1), ylim=extendrange(range(sharks$Length)),
     xlab = "Proportion p",
     ylab = bquote("Quantiles Q"["y"]*"("p)"),
     main = "US Encounters")
```

```
qvals <- sort(pop[[1]]$Length)
pvals <- ppoints(length(qvals))
plot(pvals, qvals, pch = 19, col=adjustcolor("black", alpha = 0.5),
```

```

xlim=c(0,1), ylim=extendrange(range(sharks$Length)),
xlab = "Proportion p",
ylab = bquote("Quantiles Q"["y"]*"("p)"),
main = "Australia/US Encounters")
qvals <- sort(pop[[2]]$Length)
pvals <- ppoints(length(qvals))
points(pvals, qvals, pch = 19, col=adjustcolor("red", alpha = 0.5) )

```



Comparing figures (like those above) tends to require some subjectivity. When comparing sub-populations we most often perform numerical comparisons.

- The difference between means in these sub-populations is:

$$a(\mathcal{P}_1) - a(\mathcal{P}_2) = \bar{y}_1 - \bar{y}_2$$

```
mean(pop$pop1[, "Length"]) - mean(pop$pop2[, "Length"])
```

```
## [1] 5.524436
```

- The ratio of standard deviations in these sub-populations is:

$$\frac{a(\mathcal{P}_1)}{a(\mathcal{P}_2)} = \frac{SD(\mathcal{P}_1)}{SD(\mathcal{P}_2)}$$

```
sd(pop$pop1[, "Length"]) / sd(pop$pop2[, "Length"])
```

```
## [1] 1.056418
```

Are these differences large?

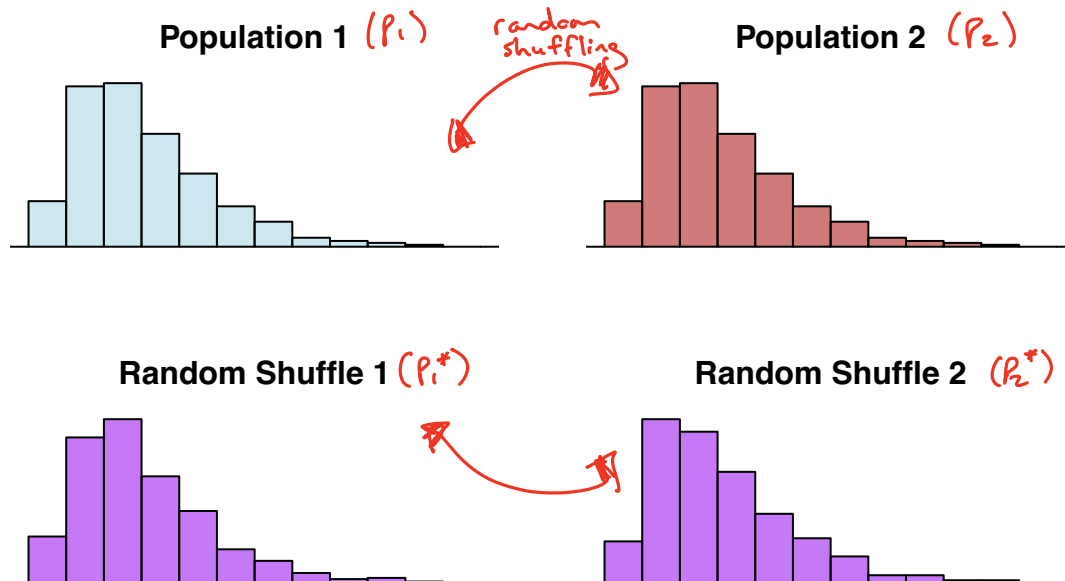
Not sure. We need to decide if the differences are much bigger than what would be expected if P_1 and P_2 really were the same.

Randomly Mixing Sub-Populations

- If the two sub-populations are essentially the same

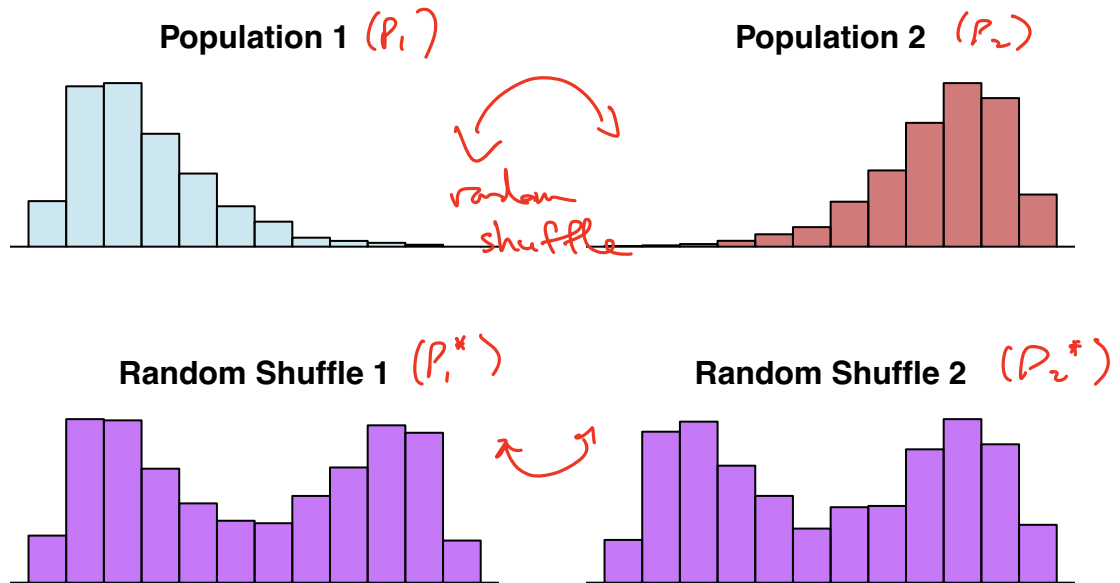
~~*~~ then the sub-populations observed should not look too different if we were to mix them up with one another

- in other words, swapping units would not dramatically change the features of the resulting sub-populations.



- On the other hand, if the two sub-populations were very different

~~*~~ then shuffling the units could dramatically change the features of the resulting sub-populations.



- Here we combine the two sub-populations together into one $\mathcal{P} = \{P_1, P_2\}$ and then

– randomly draw two new sub-populations P_1^* and P_2^*

* ensuring that the sub-population sizes are kept the same

- Here is a function that does this:

```

mixRandomly <- function(pop) {
  pop1 <- pop$pop1
  n_pop1 <- nrow(pop1)

  pop2 <- pop$pop2
  n_pop2 <- nrow(pop2)

  mix <- rbind(pop1, pop2)
  select4pop1 <- sample(1:(n_pop1 + n_pop2),
                        n_pop1,
                        replace = FALSE)

  new_pop1 <- mix[select4pop1,]
  new_pop2 <- mix[-select4pop1,]
  list(pop1=new_pop1, pop2=new_pop2)
}

```

assumes this is a list with
two components called pop1
and pop2.

1:N

n_{pop1}

replace = FALSE

P_1^*

P_2^*

returned list

- We then compare the attributes of $\{P_1, P_2\}$ with $\{P_1^*, P_2^*\}$:
 - e.g., $a(P_1)$ vs. $a(P_1^*)$ or $a(P_2)$ vs. $a(P_2^*)$
 - e.g., $a(P_1) - a(P_2)$ vs. $a(P_1^*) - a(P_2^*)$
 - e.g., $a(P_1)/a(P_2)$ vs. $a(P_1^*)/a(P_2^*)$
 - some other measure of difference among the sub-populations.

- If the sub-populations were similar to begin with, there shouldn't be a very large difference between attributes calculated on $\{P_1, P_2\}$ versus those calculated on $\{P_1^*, P_2^*\}$.

Do shark lengths differ in Australia vs. USA?

Let's re-examine the Shark example previously introduced (in which shark length was the variate of interest):

- First let's define some factory functions that will be helpful for calculating differences of means and ratios of standard deviations:

```
getAveDiffsFn <- function(variate) {
  function(pop) {mean(pop$pop1[, variate]) - mean(pop$pop2[, variate])}
}

getSDRatioFn <- function(variate) {
  function(pop) {sd(pop$pop1[, variate])/sd(pop$pop2[, variate])}
}
```

- For shark length we use the factory functions above to define the following:

```
diffAveLengths <- getAveDiffsFn("Length")
ratioSDLengths <- getSDRatioFn("Length")
```

- Recall that the difference in averages and ratio of the standard deviations between the two sub-populations are respectively:

$$\bar{y}_1 - \bar{y}_2 \quad \text{and} \quad \frac{SD(P_1)}{SD(P_2)}$$

```
round(c(diffAveLengths(pop), ratioSDLengths(pop)), 3)
```

```
## [1] 5.524 1.056
```

- Randomly mixing the two sub-populations (while maintaining $N_1 = 28$ and $N_2 = 19$) and calculating these same summaries yields

$$\bar{y}_1^* - \bar{y}_2^* \quad \text{and} \quad \frac{SD(P_1^*)}{SD(P_2^*)}$$

```
set.seed(341)
mixedPop <- mixRandomly(pop)
round(c(diffAveLengths(mixedPop), ratioSDLengths(mixedPop)), 3)

## [1] 8.528 0.969
```

- It seems that the standard deviation does not change much under shuffling, but the mean can change quite a lot.

- Are these differences unusual?

- What would have happened if our random mixing was slightly different (i.e., we obtained a *different* random partition)?
- To make this determination we need a more rigorous statistical analysis... we could perform a hypothesis test...

- We will discuss such tests from a computational perspective:

- Let's look at many such shufflings $\{\mathcal{P}_1^*, \mathcal{P}_2^*\}$
- And each time calculate $\bar{y}_1^* - \bar{y}_2^*$ and $\frac{SD(\mathcal{P}_1^*)}{SD(\mathcal{P}_2^*)}$
- Then let's make a histogram for each of these statistics and observe where $\bar{y}_1 - \bar{y}_2$ and $\frac{SD(\mathcal{P}_1)}{SD(\mathcal{P}_2)}$ lie
- There are $\binom{N_1+N_2}{N_1} = \binom{28+19}{28} = 6.97 \times 10^{12}$ potential rearrangements, but this is too many to consider. We will use 5,000 rearrangements instead.

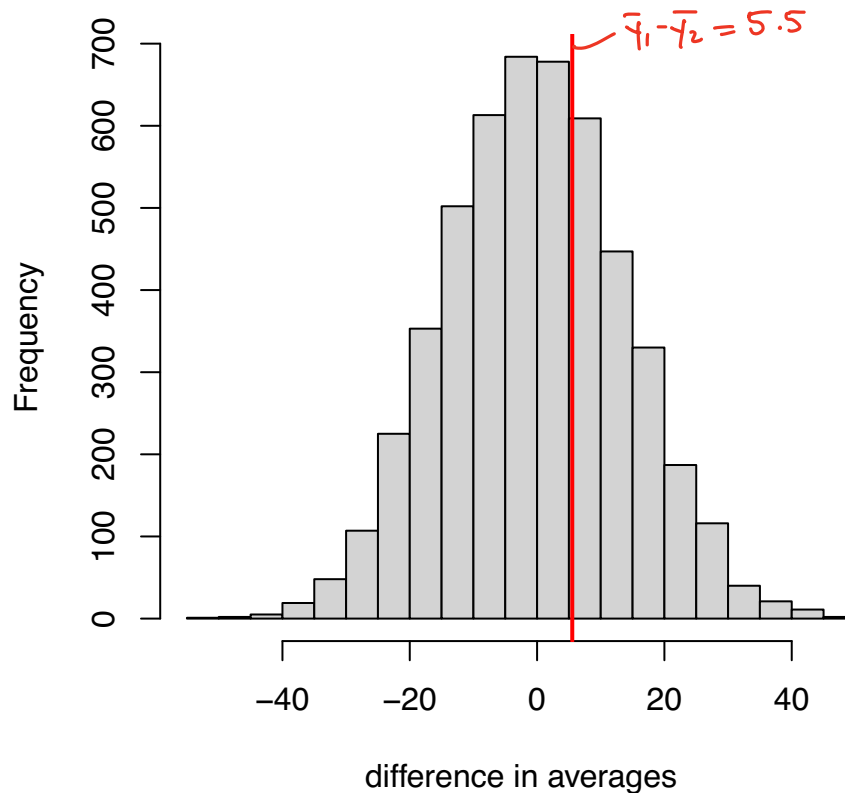
5000 values of
 $\bar{y}_1^* - \bar{y}_2^*$

- First for the **averages**:

```
set.seed(341)
diffLengths <- sapply(1:5000,
                      FUN = function(...) {diffAveLengths(mixRandomly(pop))})

hist(diffLengths, breaks=20,
     main = "Randomly Mixed Populations", xlab="difference in averages",
     col="lightgrey")
abline(v=diffAveLengths(pop), col = "red", lwd=2)
```


Randomly Mixed Populations



- The red line represents the difference between the average shark lengths in each of the sub-populations

$$a(\mathcal{P}_{Australia}) - a(\mathcal{P}_{USA})$$

- Does $a(\mathcal{P}_{Australia}) - a(\mathcal{P}_{USA})$ seem to be *extreme* relative to the randomly mixed differences $a(\mathcal{P}_1^*) - a(\mathcal{P}_2^*)$?

No, it's fairly typical

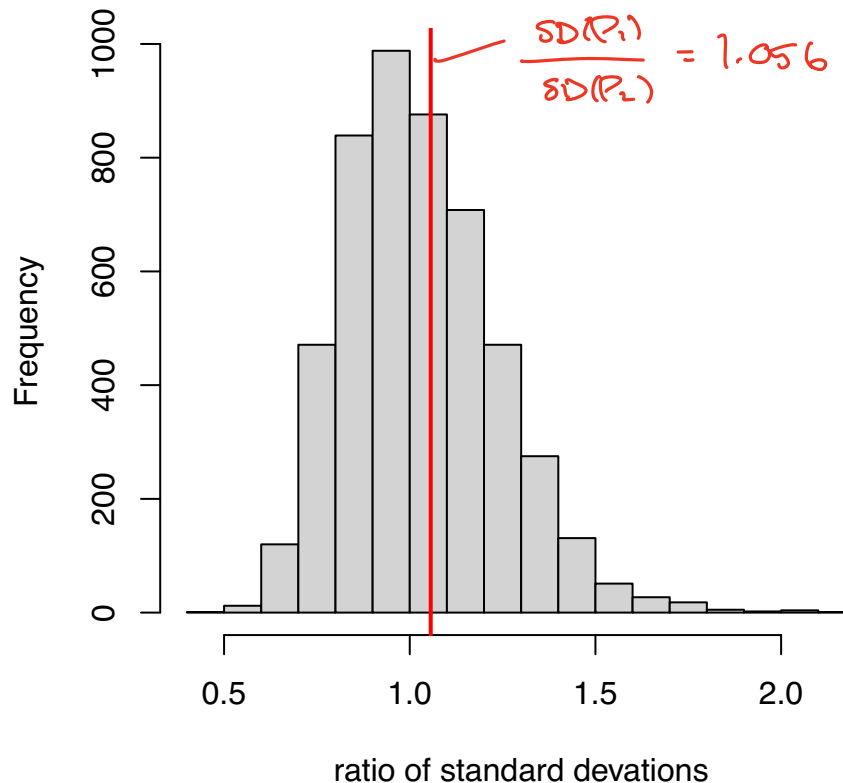
5000 values of $SD(\mathcal{P}_1^*)/SD(\mathcal{P}_2^*)$

- Next for the **standard deviations**:

```
set.seed(341)
ratioLengths <- sapply(1:5000,
  FUN = function(...){ratioSDLengths(mixRandomly(pop))})

hist(ratioLengths, breaks=20,
  main = "Randomly Mixed Populations", xlab="ratio of standard deviations",
  col="lightgrey")
abline(v=ratioSDLengths(pop), col = "red", lwd=2)
```

Randomly Mixed Populations



- The red line represents the ratio of the standard deviation of shark lengths in each of the sub-populations

$$\frac{SD(\mathcal{P}_{Australia})}{SD(\mathcal{P}_{USA})}$$

- Does $SD(\mathcal{P}_{Australia})/SD(\mathcal{P}_{USA})$ seem to be *extreme* relative to the randomly mixed ratios $SD(\mathcal{P}_1^*)/SD(\mathcal{P}_2^*)$?

No, this is fairly typical.

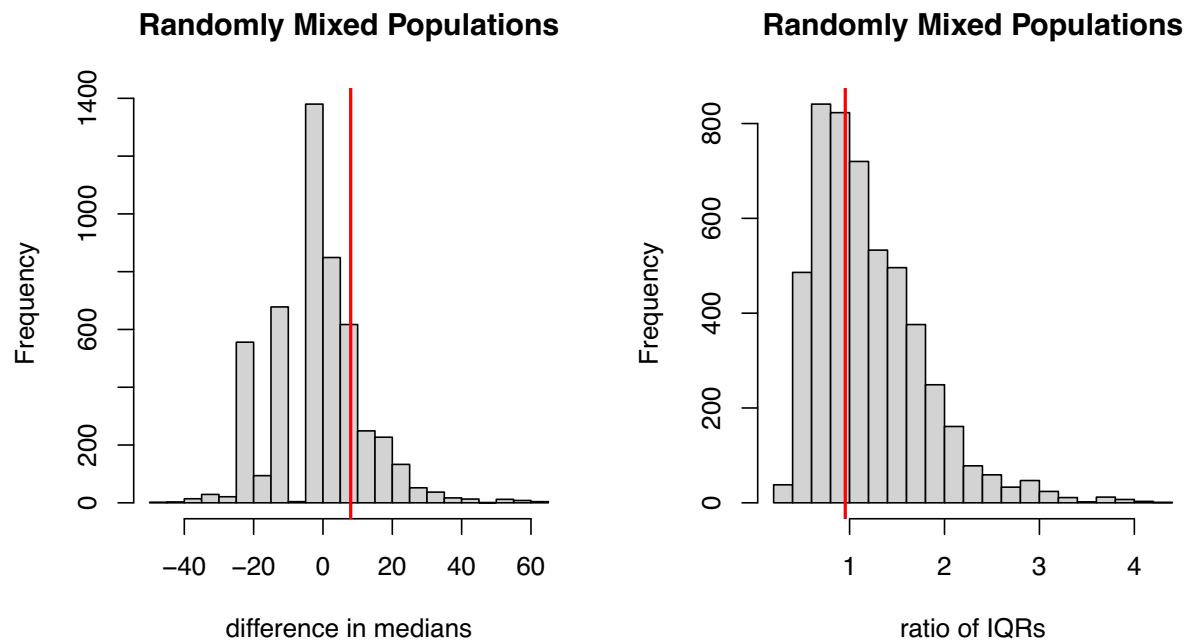
- We can also compare these sub-populations on the basis of their medians and interquartile ranges.
 - But first let's define some helpful functions like we did before:

```
getMedianDiffsFn <- function(variate) {
  function(pop) {median(pop$pop1[, variate]) - median(pop$pop2[, variate])}
}

getIQRRatioFn <- function(variate) {
  function(pop) {IQR(pop$pop1[, variate])/IQR(pop$pop2[, variate])}
}
```

```
diffMedianLengths <- getMedianDiffsFn("Length")
ratioIQRLengths <- getIQRRatioFn("Length")
```

- Now, randomly mix the populations and summarize the results with a histogram.



- In both cases the red line represents the discrepancy measure calculated on the observed sub-populations $\{\mathcal{P}_{Australia}, \mathcal{P}_{USA}\}$

- What have we learned from these plots?

It appears as though shark lengths in Australia vs. the US are comparable. BUT this conclusion is based specifically on a comparison of means, medians, IQR's, and SD's.

Do shark lengths differ in fatal vs. non-fatal encounters?

- The two sub-populations are $\mathcal{P}_1 = \text{Fatal shark encounters}$ and $\mathcal{P}_2 = \text{Non-Fatal shark encounters}$
 - And the variate of interest is again shark length
 - The code below constructs the two sub-populations:

```
Fatpop <- list(pop1 = sharks[sharks[, "Fatality"] == 1, ],
               pop2 = sharks[sharks[, "Fatality"] == 0, ])
```

- We can numerically compare the sharks lengths from the two populations:

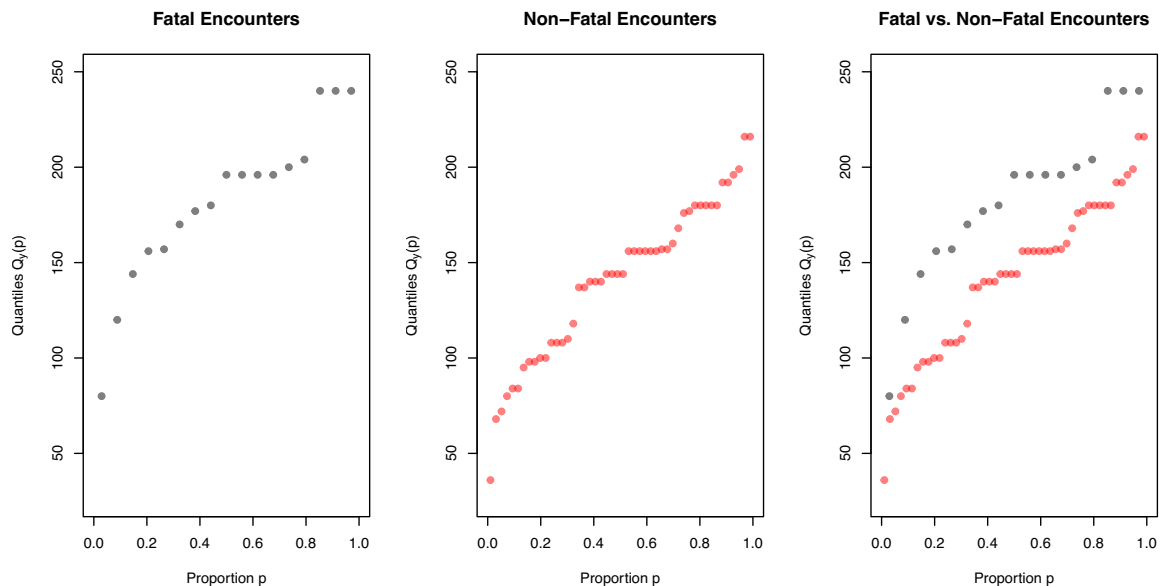
```
summary(Fatpop[[1]]$Length)
```

```
##      Min. 1st Qu.  Median    Mean 3rd Qu.    Max.
##      80.0  157.0   196.0   181.9   200.0   240.0
```

```
summary(Fatpop[[2]]$Length)
```

```
##      Min. 1st Qu.  Median    Mean 3rd Qu.    Max.
##      36.0  108.0   144.0   141.2   176.2   216.0
```

- Let's also graphically compare the sharks lengths from the two populations by way of quantile plots:



- It seems that fatal encounters involve bigger sharks compared to non-fatal encounters. (Why?)
- For shark encounters involving fatal and non-fatal encounters we quantify the difference in the average and standard deviation of the shark lengths from the two sub-populations by randomly mixing the sub-populations.

```
Fatpop <- list(pop1 = sharks[sharks[, "Fatality"] == 1, ],
              pop2 = sharks[sharks[, "Fatality"] == 0, ])

par(mfrow=c(1,2), oma=c(0,0,2,0))

set.seed(341)
fatpair <- sapply(1:5000,
  FUN = function(...){
    tmixpop = mixRandomly(Fatpop)
```

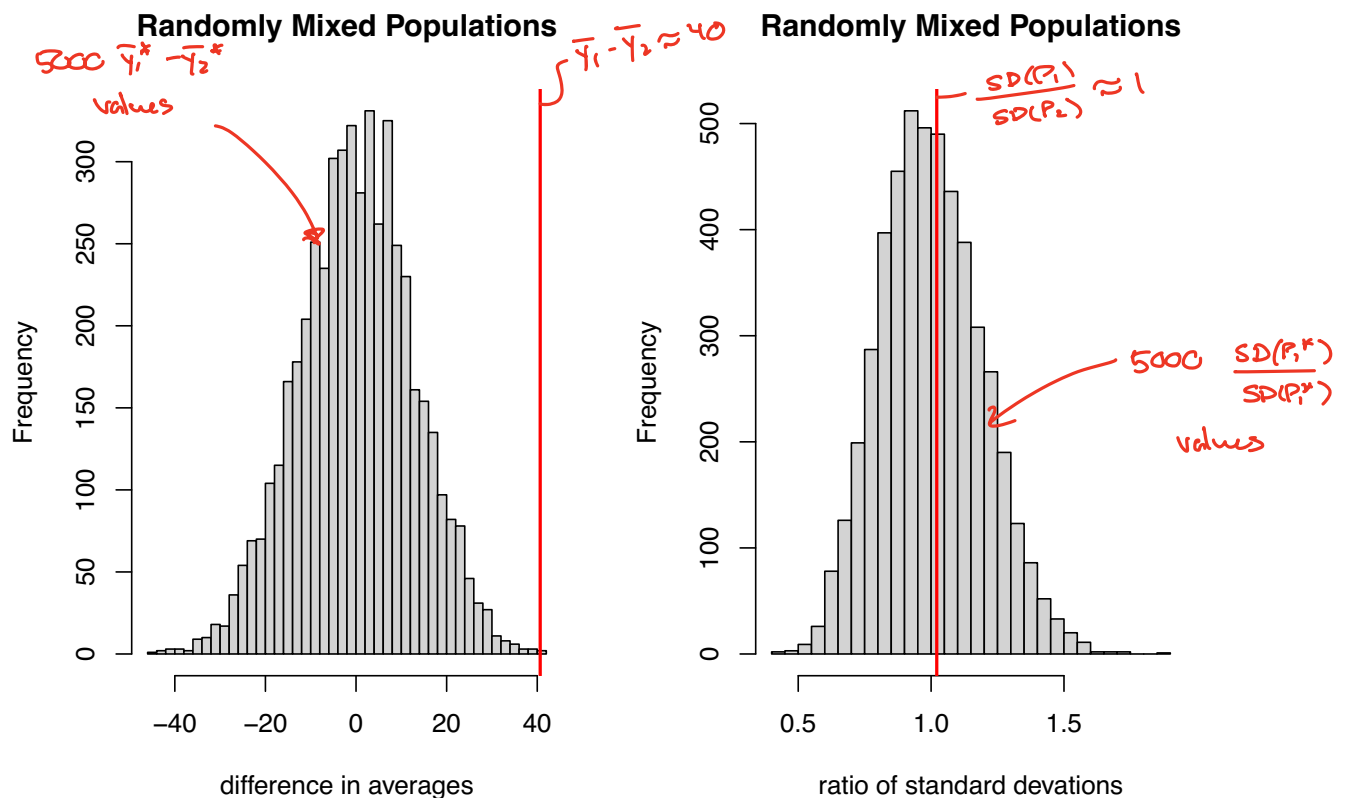
```

c( diffAveLengths(tmixpop), ratioSDLenghts(tmixpop))  })

hist(fatpair[1,], breaks="FD",
     main = "Randomly Mixed Populations", xlab="difference in averages",
     col="lightgrey")
abline(v=diffAveLengths(Fatpop), col = "red", lwd=2)

hist(fatpair[2,], breaks="FD",
     main = "Randomly Mixed Populations", xlab="ratio of standard deviations",
     col="lightgrey")
abline(v=ratioSDLenghts(Fatpop), col = "red", lwd=2)

```



- What do these plots say about the lengths of sharks involved in fatal vs. non-fatal encounters?

We see a substantial difference in average shark lengths, but the variability of shark lengths is similar between fatal and non-fatal encounters.

Summary

- We motivated comparing sub-populations using the shark encounter data.
- We assess whether differences between observed sub-populations are extreme, relative to differences between randomly mixed/ shuffled sub-populations.

Exercises

- From the Inference exercises try problem 4 parts (a) and (b).

4.2.1 Anatomy of a Significance Test

Overview

- We break down the steps of a Significance Test and consider an example.
-

Introduction

We would like to quantify, numerically, how unusual the difference between $a(\mathcal{P}_1)$ and $a(\mathcal{P}_2)$ is relative to randomly mixed sub-populations.

- * If the two sub-populations are actually similar, we want to provide numerical evidence in favour of the notion that the two sub-populations are similar to randomly mixed sub-populations.
- * If the two sub-populations are actually different, we want to provide numerical evidence against the notion that the two sub-populations are similar to randomly mixed sub-populations.

The following steps are used to gather such evidence:

1. We suppose the sub-populations were randomly drawn from the same population. This is known as the null hypothesis (H_0)

[3.5] To evaluate the extremity of the observed discrepancy we compare its value to an approximation of the sampling distribution. We obtain this approximation through¹⁴ repeated random shuffling. The comparison is formalized with a p-value.

- AKA "test statistic"
2. We construct a **discrepancy measure** that quantifies how inconsistent our data is with the null hypothesis
 - where *large values give evidence against H_0 .*
 3. We obtain the **observed discrepancy** by calculating
 - the discrepancy measure on the *two observed (i.e., unshuffled) sub-populations*
 4. Finally, we obtain the **observed p-value** by calculating
 - * the probability that a randomly shuffled sub-population has a discrepancy measure at least as large as the observed discrepancy
 - where *small values give evidence against H_0 .*

Let's elaborate on each of these steps.

The Null Hypothesis

- Each of the following (equivalent) statements constitutes the null hypothesis we are testing.
 - H_0 : The sub-populations $\mathcal{P}_1$ and $\mathcal{P}_2$ were *randomly drawn from the same population*
 - H_0 : $\mathcal{P}_1$ and $\mathcal{P}_2$ were created by *randomly assigning units from the same population to one of two sub-populations*.
 - H_0 : $\mathcal{P}_1$ and $\mathcal{P}_2$ were *generated by random mixing*
 - H_0 : $\mathcal{P}_1$ and $\mathcal{P}_2$ are *indistinguishable*.

* Regardless of how the null hypothesis is stated, the **alternative hypothesis H_A** is the complement of H_0 .

H_A : $\mathcal{P}_1$ and $\mathcal{P}_2$ are distinguishable

- In the context of the shark data, we have
 - H_0 : $\mathcal{P}_{Australia}$ and $\mathcal{P}_{USA}$ are drawn from the same population of shark encounters.
 - H_A : $\mathcal{P}_{Australia}$ and $\mathcal{P}_{USA}$ are **not** drawn from the same population of shark encounters.

Note that we do not state the null hypothesis in terms of the equivalence of attribute values, i.e. $a(\mathcal{P}_1) = a(\mathcal{P}_2)$.

H_0 : $a(\mathcal{P}_1) = a(\mathcal{P}_2)$ vs. H_A : $a(\mathcal{P}_1) \neq a(\mathcal{P}_2)$

* Although such a statement is true if H_0 holds, it is *weaker* and so we avoid using it.

The Discrepancy Measure

- A discrepancy measure $D(\mathcal{P}_1, \mathcal{P}_2)$ quantifies how inconsistent our data is with the null hypothesis, and is defined so that large values indicate evidence against the null hypothesis.

– As a point of interest, the discrepancy measure is technically an attribute for the population $\mathcal{P} = \{\mathcal{P}_1, \mathcal{P}_2\}$ and so we could consider properties such as equivariance and invariance.

* In other statistical texts the discrepancy measure is often referred to as a **test statistic**

positive

- The form of $D(\mathcal{P}_1, \mathcal{P}_2)$ depends on how we want to compare $\mathcal{P}_1$ and $\mathcal{P}_2$

– If we want to compare measures of location the discrepancy measure is typically based on *differences*:

$$D(\mathcal{P}_1, \mathcal{P}_2) = \mu(\mathcal{P}_1) - \mu(\mathcal{P}_2)$$

– If we want to compare measures of spread the discrepancy measure is typically based on *ratios*:

$$D(\mathcal{P}_1, \mathcal{P}_2) = \frac{\sigma(\mathcal{P}_1)}{\sigma(\mathcal{P}_2)}$$

- For example:

– if we hypothesized that the averages from the two sub-populations were the same, then a discrepancy measure for this might be

$$D(\mathcal{P}_1, \mathcal{P}_2) = |\bar{y}_1 - \bar{y}_2|$$

– if we hypothesized that the standard deviation from the two sub-populations were the same, then a discrepancy measure for this might be

$$D(\mathcal{P}_1, \mathcal{P}_2) = \left| \frac{SD(\mathcal{P}_1)}{SD(\mathcal{P}_2)} - 1 \right|$$

– if we hypothesized that the average from the first population was smaller than the average from the second population, then a discrepancy measure for this might be

Consistent with H_0
 $\bar{y}_1 < \bar{y}_2 \rightarrow \bar{y}_1 - \bar{y}_2 < 0$

$$D(\mathcal{P}_1, \mathcal{P}_2) = \bar{y}_1 - \bar{y}_2$$

– if we hypothesized that the average from the first population was larger than the average from the second population, then a discrepancy measure for this might be

$$\bar{y}_1 > \bar{y}_2 \rightarrow \bar{y}_1 - \bar{y}_2 > 0$$

↑
 Consistent with H_0

$$D(\mathcal{P}_1, \mathcal{P}_2) = \bar{y}_2 - \bar{y}_1$$

The Observed Discrepancy

- The observed discrepancy, d_{obs} , is the value of discrepancy measure D calculated on the two observed (i.e., unshuffled) sub-populations:

$$d_{obs} = D(P_1, P_2) \quad \text{original unshuffled data}$$

- It's important to recognize that the discrepancy measure quantifies only one type of discrepancy between the populations
 - e.g., discrepancy in averages
 - e.g., discrepancy in standard deviations

* All other differences are completely ignored.

For instance P_1 and P_2 could be very similar with respect to their means, but very different with respect to their variation.

probability of observing a discrepancy at least as extreme as the one we observed if H_0 was true.

The Observed p-value

- The observed p -value is the probability that a randomly shuffled sub-population has a discrepancy measure at least as large as the observed discrepancy

$$p\text{-value} = \Pr(D \geq d_{obs} \mid H_0 \text{ is true})$$

"at least as extreme"

sampling distribution of D values approximated via random shuffling (i.e., the null distribution)

- If the p -value is very small then either
 - the null hypothesis is true and we have observed a very unusual value of d_{obs}
 - OR the null hypothesis is false.

This is unlikely, so we choose to believe this

- The smaller the p -value, the greater the evidence against the null hypothesis.
 - $p\text{-value} < 0.001$ means that there is **very strong evidence** against H_0
 - $0.001 < p\text{-value} < 0.01$ means that there is **strong evidence** against H_0
 - $0.01 < p\text{-value} < 0.05$ means that there is **evidence** against H_0
 - $0.05 < p\text{-value} < 0.1$ means that there is **weak evidence** against H_0
 - $p\text{-value} > 0.1$ means that there is **no evidence** against H_0
 - In the extreme case where $p\text{-value} = 0$, then we have observed something impossible and the hypothesis must therefore be false – this would be a proof by contradiction.

"Permutation Test"

- In order to calculate the p-value exactly one must consider all $\binom{N_1+N_2}{N_1} = \binom{N_1+N_2}{N_2}$ possible permutations of the observed data

– the exact p-value is the proportion of $D(P_1^*, P_2^*)$ values are greater than or equal to d_{obs} .

- Because $\binom{N_1+N_2}{N_1} = \binom{N_1+N_2}{N_2}$ is in practice too many permutations to consider, we typically just use M (a large number) of them

– In particular we generate M shuffled pairs:

$$(P_{1,1}^*, P_{2,1}^*), (P_{1,2}^*, P_{2,2}^*), \dots, (P_{1,M}^*, P_{2,M}^*)$$

– The p-value is then approximated as

$$\hat{p}\text{-value} = \frac{1}{M} \sum_{i=1}^M \mathbb{I}[D(P_{1,i}^*, P_{2,i}^*) \geq d_{obs}]$$

"Randomization Test"

- If $M = \binom{N_1+N_2}{N_1} = \binom{N_1+N_2}{N_2}$ and we've considered all possible shuffles the equation above would yield the exact p-value

Putting it All Together

Test of Significance Algorithm

1. State the **null hypothesis**: $H_0: \mathcal{P}_1$ and $\mathcal{P}_2$ are drawn from the same population.
2. Construct a measure of **discrepancy** $D = D(\mathcal{P}_1, \mathcal{P}_2)$ where large values indicate **evidence against the null hypothesis**
3. Calculate the **observed discrepancy** $d_{obs} = D(\mathcal{P}_1, \mathcal{P}_2)$.
4. Shuffle the sub-populations M times and calculate the **observed p-value**:

$$p\text{-value} = Pr(D \geq d_{obs} \mid H_0 \text{ is true}) \approx \frac{1}{M} \sum_{i=1}^M \mathbb{I}(D(\mathcal{P}_{1,i}^*, \mathcal{P}_{2,i}^*) \geq d_{obs})$$

Shark Example

- First let's define some factory functions that will be helpful for calculating differences of means and ratios of standard deviations:

```
getAveDiffsFn <- function(variate) {
  function(pop) {mean(pop$pop1[, variate]) - mean(pop$pop2[, variate])}
}

getSDRatioFn <- function(variate) {
  function(pop) {sd(pop$pop1[, variate])/sd(pop$pop2[, variate])}
}
```

- For shark length we use the factory functions above to define the following:

```
diffAveLengths <- getAveDiffsFn("Length")
ratioSDLenghts <- getSDRatioFn("Length")
```

- Recall that the difference in averages and ratio of the standard deviations between the two sub-populations are respectively:

$$\bar{y}_1 - \bar{y}_2 \quad \text{and} \quad \frac{SD(\mathcal{P}_1)}{SD(\mathcal{P}_2)}$$

```
round(c(diffAveLengths(pop), ratioSDLenghts(pop)), 3)
```

```
## [1] 5.524 1.056
```

- Randomly mixing the two sub-populations (while maintaining $N_1 = 28$ and $N_2 = 19$) and calculating these same summaries yields

$$\bar{y}_1^* - \bar{y}_2^* \quad \text{and} \quad \frac{SD(\mathcal{P}_1^*)}{SD(\mathcal{P}_2^*)}$$

```
set.seed(341)
mixedPop <- mixRandomly(pop)

diffLengths <- sapply(1:5000,
  FUN = function(...){diffAveLengths(mixRandomly(pop))})
round(c(diffAveLengths(mixedPop), ratioSDLenghts(mixedPop)), 3)
```

```
## [1] 8.528 0.969
```

- In R, since we already have a sample of $M = 5000$ shuffled pairs, we can calculate the approximate p -value with `sum(abs(diffLengths) >= abs(diffAveLengths(pop))) / length(diffLengths)`

– Note that this assumes a discrepancy measure of $D(\mathcal{P}_1, \mathcal{P}_2) = |\bar{y}_1 - \bar{y}_2|$

- For the null hypothesis

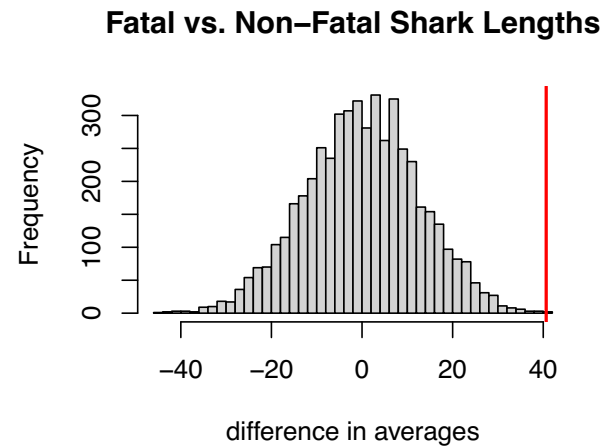
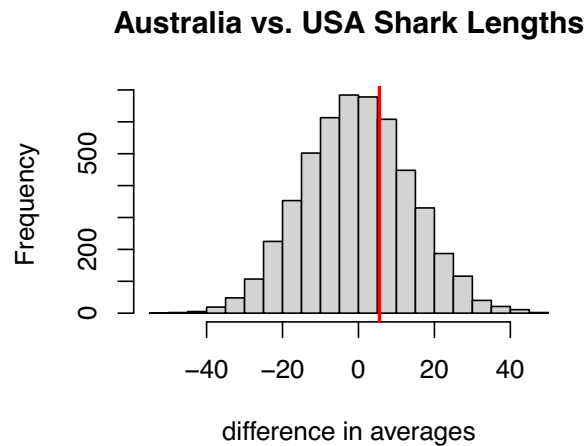
✱ $H_0 : \mathcal{P}_{Australia} \text{ and } \mathcal{P}_{USA} \text{ are drawn from the same population of shark lengths}$

we find $p\text{-value} \approx 0.7014$

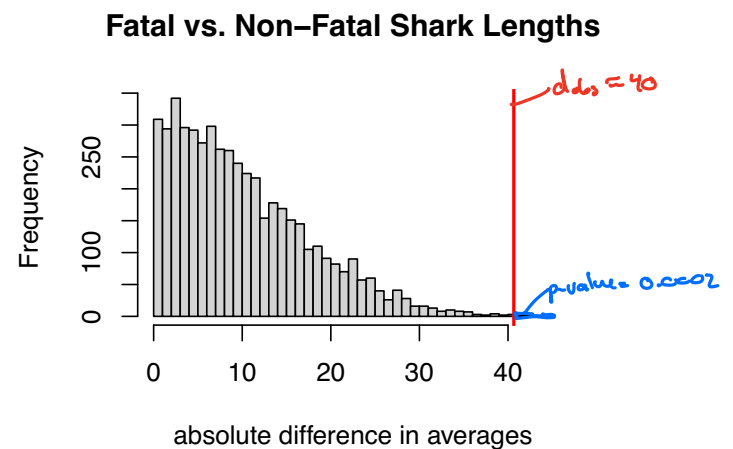
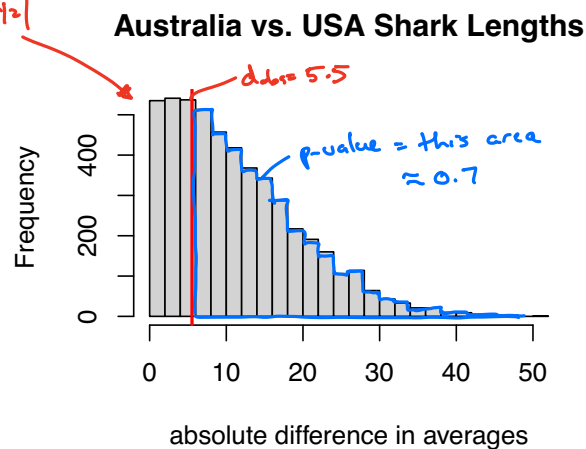
- For the null hypothesis

$H_0 : \mathcal{P}_{Fatal}$ and $\mathcal{P}_{Non-Fatal}$ are drawn from the same population of shark lengths

we find $p\text{-value} \approx 0.0002$



Sampling dist of
 $|\bar{y}_1 - \bar{y}_2|$



* Suppose that the pair $(\mathcal{P}_{Australia}, \mathcal{P}_{USA})$ is a random draw:

- then the probability of seeing at least as large a difference as we observed in $(\mathcal{P}_{Australia}, \mathcal{P}_{USA})$ is approximately 0.7014
- thus, there is **no evidence against the null hypothesis** that the pair $(\mathcal{P}_{Australia}, \mathcal{P}_{USA})$ was randomly drawn

- in other words, we have no evidence against the hypothesis that the two populations $\mathcal{P}_{Australia}$ and $\mathcal{P}_{USA}$ are indistinguishable

✖ Suppose that the pair $(\mathcal{P}_{Fatal}, \mathcal{P}_{Non-Fatal})$ is a random draw:

- then the probability of seeing at least as large a difference as we observed in $(\mathcal{P}_{Fatal}, \mathcal{P}_{Non-Fatal})$ is approximately 0.0002
- thus, there is **very strong evidence against the null hypothesis** that the pair $(\mathcal{P}_{Fatal}, \mathcal{P}_{Non-Fatal})$ was randomly drawn
- in other words, we have very strong evidence against the hypothesis that the two populations $\mathcal{P}_{Fatal}$ and $\mathcal{P}_{Non-Fatal}$ are indistinguishable

Significance Testing Errors

Courtroom Analogy

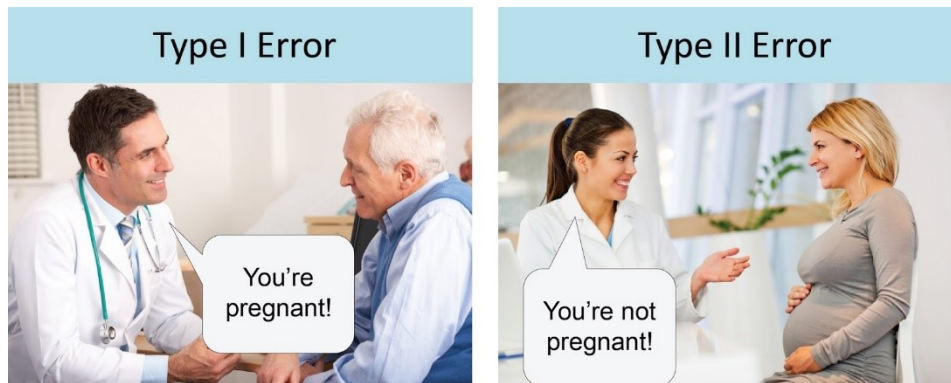
Decision	defendant is innocent	the defendant is guilty
Convicted	Error (Type I Error)	Correct
Acquitted	Correct	Error (Type II Error)

In Hypothesis Testing

Decision	H_0 is true	H_0 is false
Reject H_0	Error (Type I Error) ☹️	Correct 😊
Do Not Reject H_0	Correct 😊	Error (Type II Error) ☹️

Another Example

H_0 : person is not pregnant vs. H_A : person is pregnant



A note on the language:

- We will try to avoid using terms such as “reject”, “fail to reject”, “accept”, etc.
- Instead, we use the significance test simply to measure the evidence against H_0

Important Remarks

- The observed p -value provides a common (probabilistic) scale on which to measure the **evidence against the null hypothesis**

* The observed p -value does **not** measure evidence in favour of the null hypothesis.

- in science, we try to falsify hypotheses and entertain only those which remain standing;

Absence of evidence $\neq$ evidence of absence

- A test of significance therefore **neither accepts nor rejects a null hypothesis**; it simply provides a measure of the evidence against it

- the decision taken in light of this evidence is the choice of the researcher

- There is **no magic level** for a p -value such as 0.05 or 0.01,

- there is no practical or scientific difference between p -value = 0.048 and p -value = 0.051, for example

* The fact that the evidence against the null hypothesis is **statistically significant** based on some discrepancy measure **does not imply that the discrepancy is practically significant**

- the p -value measures how unusual a discrepancy of that size might be when the null hypothesis holds,
- it says nothing about whether a discrepancy of that size matters for any practical or scientific purpose
- e.g., for the shark lengths data, the average shark length in fatal vs. non-fatal encounters differed by 3.25 feet. Is that difference of practical importance?

* Every test of significance is based on some measure of discrepancy and **different discrepancy measures can detect different departures** from the null hypothesis, so one needs to understand the nature of the departure from the hypothesis that the discrepancy is trying to measure.

See: Equivalence and non-inferiority test

Summary

- We investigated each step of a Significance Test
- We use the shark encounter data as an example for a Significance Test
- We reviewed possible significance test errors and some important remarks.

Exercises

- From the Inference exercises try problem 8 parts (a), (b), (c).

4.2.2 A t-like Discrepancy Measure

This content is covered in Tutorial 9. See those notes.

Overview

- We investigate a special discrepancy measure that looks like the statistic from a t-test.
-

Introduction

- When comparing two sub-populations on the basis of a measure of location, one particularly useful discrepancy measure is
- This discrepancy measure is “physically dimensionless”
 - Whatever scale the numerator is measured in (e.g. inches as in the shark lengths), the scale of the denominator will match, leaving the ratio free of any measurement scale.
 - This naturally makes this discrepancy measure
- The challenge is determining the denominator of the discrepancy measure.
 - In rare cases, the denominator might be known and then this discrepancy measure is a rescaling of $a(\mathcal{P}_1) - a(\mathcal{P}_2)$ and would not yield different results.
 - However, more commonly, we will estimate the denominator using information from $\mathcal{P}_1$ and $\mathcal{P}_2$.

- Suppose that the populations $\mathcal{P}_1$ and $\mathcal{P}_2$ are **random** and **independently** drawn from the **same** larger population. Then the denominator should be instead

- But determining the form of $SD[\tilde{a}(\mathcal{P}_1) - \tilde{a}(\mathcal{P}_2)]$ can also be difficult
 - Except in the common special case when $a(\mathcal{P})$ is an average

Differences in Averages

Suppose we were interested in differences in averages:

- In this case $a(\mathcal{P}_i) = \bar{Y}_i$ and $\mathcal{P}_i$ has size N_i , $i = 1, 2$
- And the discrepancy measure becomes:

where $\tilde{\sigma}$ is an estimator of the standard deviation of the Y values in the population $\mathcal{P} = \{\mathcal{P}_1, \mathcal{P}_2\}$.

- If $\tilde{\sigma}_1$ and $\tilde{\sigma}_2$ denote the estimators of the standard deviations from each of $\mathcal{P}_1$ and $\mathcal{P}_2$ respectively, then the pooled estimator of σ would be

- **Note:** if it were inappropriate to assume the variability in the two sub-populations was equivalent we could instead use the denominator

This just looks like the t-test

- This *is* the “two-sample” Student t statistic used to test the equality of the means of two normal distributions with common (but unknown) standard deviation σ .
 - If the Y values were in fact normally distributed, the discrepancy measure would follow a Student t distribution with $N_1 + N_2 - 2$ degrees of freedom under the null hypothesis that the means were identical.
- Note, however, in our procedure of randomly mixing the populations we make **no such normality assumption**.
 - We simply proceed with *this* discrepancy measure just as we did with the earlier measures. The only difference is that now we need to first calculate the denominator (the standard error).
- Below is a factory function that will return a function that calculates this discrepancy measure for any two sub-populations for a given variate **var** (assuming the variance of the sub-populations are equal)

```
### The t statistic

getDiscrepancyFn <- function(var) {
  function(pop) {
    ## First sub-population
    pop1 <- pop$pop1
    n1 <- nrow(pop1)
    m1 <- mean(pop1[, var])
    v1 <- var(pop1[, var])

    ## Second sub-population
    pop2 <- pop$pop2
    n2 <- nrow(pop2)
    m2 <- mean(pop2[, var])
    v2 <- var(pop2[, var])

    ## Pool the variances
    v <- ((n1 - 1) * v1 + (n2 - 1) * v2) / (n1 + n2 - 2)

    ## Determine the t-statistic
    t <- (m1 - m2) / sqrt(v * ( (1/n1) + (1/n2) ) )

    ## Return the t-value
    t
  }
}
```

Comparing Shark Lengths in Australia vs. USA

- Get the t -like discrepancy measure for the variate `Length`

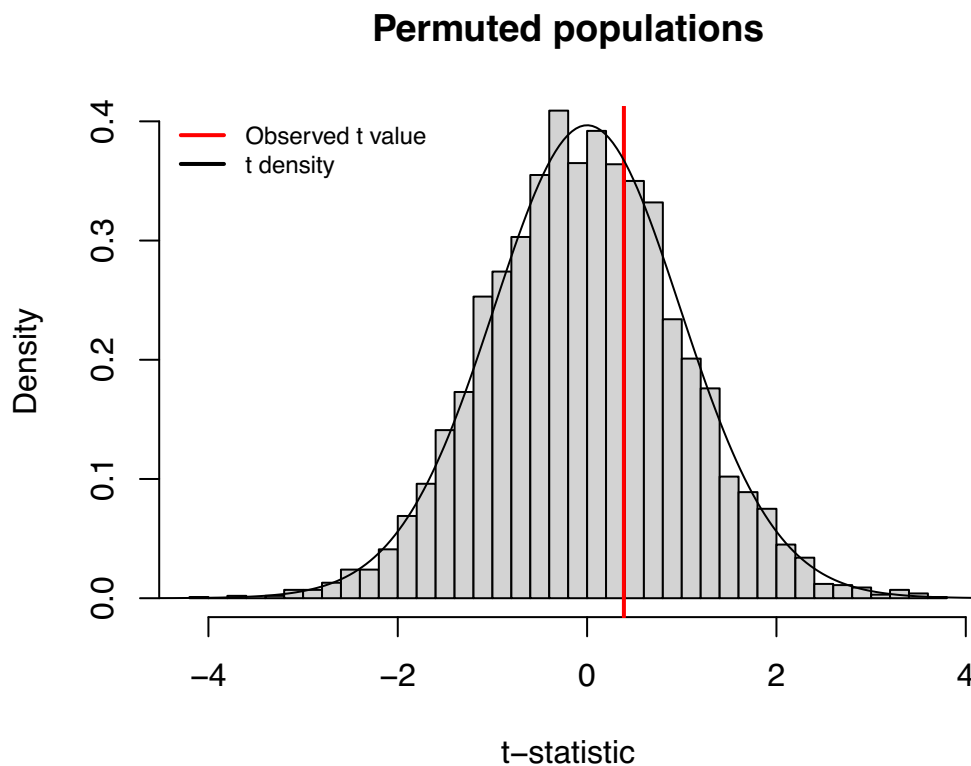
```
tStatLengths <- getDiscrepancyFn("Length")
```

- The t -like discrepancy measure calculated on the Australia vs. USA sub-populations is

```
sharks <- read.csv("../Data/sharks.csv",  
                  header = TRUE)  
  
pop <- list(pop1 = subset(sharks, sharks$Australia == 1),  
            pop2 = subset(sharks, sharks$USA == 1))  
  
tStatLengths(pop)
```

```
## [1] 0.3886752
```

- To gauge the size of the this discrepancy measure we
 - mix / shuffle / permute the sub-populations $M = 5000$ times and plot the histogram as before
 - for comparison we also overlay the Student t density (with $N_1 + N_2 - 2$ degrees of freedom) which is what we would use to calculate p -values if we were willing to make an assumption of normality



- Remarkably, the Student t density closely approximates the histogram!
 - In many instances, even when no normality distribution is assumed, the Student t distribution will roughly approximate the histogram that arises from randomly mixing the sub-populations.
 - This in fact was one of the early justifications (by R.A. Fisher) for using the t distribution broadly in application; namely that it approximated the random mixing procedure.

- The p -value can now be estimated two ways:

1) Using the sub-populations generated via random mixing:

```
tobs = tStatLengths(list(pop1 = sharks[sharks[, "Australia"] == 1, ], pop2 = sharks[sharks[, "USA"] == 1, ])
mean(abs(tVals) >= abs(tobs) )
```

```
## [1] 0.6998
```

2) Using the t -distribution with $N_1 + N_2 - 2$ degrees of freedom:

```
2*pt( abs(tobs), df = (n1 + n2 - 2), lower.tail=FALSE)
```

```
## [1] 0.6993494
```

- The similarity of these two numbers shows that, in this particular example, the t -distribution could be a good approximation (though we are not using it!)
- The p -value is so large that the observed discrepancy measure is not at all unusual when the hypothesis is true.
 - This test provides **no evidence against the null hypothesis**.
 - From a practical point of view, this means that the two sub-populations Australia and USA are not different from the perspective of average shark length.

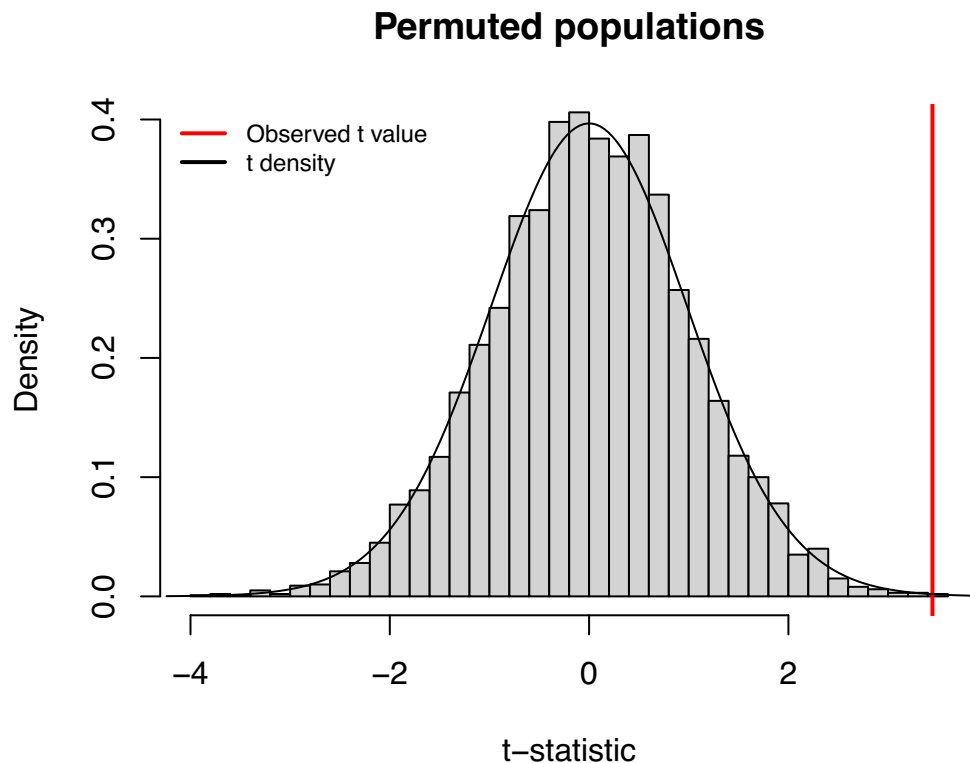
Comparing Shark Lengths in Fatal vs. Non-Fatal Encounters

- The t -like discrepancy measure calculated on the Fatal vs. Non-Fatal sub-populations is

```
Fatpop <- list(pop1 = sharks[sharks[, "Fatality"] == 1, ],
               pop2 = sharks[sharks[, "Fatality"] == 0, ])
tStatLengths(Fatpop)
```

```
## [1] 3.445492
```

- To gauge the size of the this discrepancy measure we
 - mix / shuffle / permute the sub-populations $M = 5000$ times and plot the histogram as before
 - for comparison we also overlay the Student t density (with $N_1 + N_2 - 2$ degrees of freedom) which is what we would use to calculate p -values if we were willing to make an assumption of normality



- Again, the Student t density appears to closely approximate the histogram.

- As before, the p -value can now be estimated two ways:

1) Using the sub-populations generated via random mixing:

```
tobs = tStatLengths(list(pop1 = sharks[sharks[, "Fatality"] == 1, ], pop2 = sharks[sharks[, "Fatality"] == 0, ])
SL.hat = mean(abs(tVals) >= abs(tobs))
sprintf("%.4f", SL.hat )
```

```
## [1] "0.0010"
```

2) Using the t -distribution with $N_1 + N_2 - 2$ degrees of freedom:

```
SL.that = 2*pt( abs(tobs), df = (n1 + n2 - 2), lower.tail=FALSE)
sprintf("%.5f", SL.that )
```

```
## [1] "0.00125"
```

- The observed p -value is very small, indicating that the observed discrepancy measure is unusual when the hypothesis is true.
 - This test provides **evidence against the null hypothesis**.
 - From a practical point of view, this means that the two sub-populations (Fatal vs. Non-Fatal encounters) seem to be different from the perspective of average shark length.

Summary

- We motivate a discrepancy measure that looks like t-test statistics but here we do not make a normality assumption.
- We compared the randomly mixed sub-populations associated with the t-like discrepancy to the t-distribution.
- We illustrated this t-like discrepancy measure with an example.

Exercises

- From the Inference Exercises try problem 4.

This section will not be covered in the Fall 2023 offering of STAT 341 but annotations are included for anyone interested in self study.

4.2.3 Multiple Testing

Overview

- Here we consider how to combine several discrepancy measures into a single test of significance.
-

Introduction

The null hypothesis we have been testing is the following:

H_0 : The sub-populations $\mathcal{P}_1$ and $\mathcal{P}_2$ were randomly drawn from the same population

Or, in other words:

H_0 : The sub-populations *are indistinguishable.*

- We have tested this hypothesis using a variety of discrepancy measures:

- Comparisons of means
- Comparisons of medians
- Comparisons of standard deviations
- Comparisons of interquartile ranges

;

- In general, to compare sub-populations we could use any number of discrepancy measures $D_1, D_2, \dots, D_K$

- Each with an associated p -value: $p_1, p_2, \dots, p_K$

- However, when a *family of statistical inferences* is considered simultaneously one encounters the

- The more inferences made, the more likely an error is going to occur
- e.g., Even if H_0 is true, the more discrepancy measures (and hence tests) we consider, the more likely it becomes that one of them will erroneously suggest that null hypothesis should be rejected.

** Type I error: Reject true H_0*

- If a single test has probability α of yielding a Type I Error, then this probability becomes inflated when considering K simultaneous tests

- Such an inflation is commonly quantified by two metrics:

- The **family-wise error rate (FWER)** is the probability of making a Type I Error on *any* of the K tests.
- The **false discovery rate (FDR)** is the expected number of Type I Errors in K tests.

→ FWER = $\Pr(\text{Type I error in any of the } K \text{ tests})$

• FWER $\gg \alpha$

• $\lim_{K \rightarrow \infty} \text{FWER} = 1$

→ FDR = $E[\text{Number of Type I errors in } K \text{ tests}]$

- When multiple testing cannot be avoided, many statistical methods have been developed to control FWER and FDR at acceptable values:

- Bonferroni Correction
- Sidak Correction
- Holm-Bonferroni Method
- Benjamini-Hochberg Procedure
- ...

- However, we have been considering a special case in which we are testing the same hypothesis using different discrepancy values.

- And so we can employ a more tailored solution which simply *combines* the information gained by each of the K discrepancy measures rather than considering them in isolation.

Combining Information Across Tests

- To consider the p -values collectively we might consider the smallest of them as measuring the combined evidence against the null hypothesis, i.e.

$$p\text{-value}_{\min} = \min_{k=1,2,\dots,K} p_k$$

- * Note that this is appropriate only because the p -values are on a common (interpretable) scale, i.e., they're probabilities
- * We could not, for instance, combine discrepancy measures in the same way

- The smaller the value of $p\text{-value}_{\min}$ the greater is the evidence against the null hypothesis.

- * Note: $p\text{-value}_{\min}$ is **not** a p -value in the traditional sense

- but it is a measure of the evidence against the hypothesis.

- Thus we can construct a discrepancy measure out of it:

$$D^* = 1 - p\text{-value}_{\min}$$

- D^* is defined so that large values, again, indicate evidence against the null hypothesis (unlike the p -value)
- Therefore, D^* is a discrepancy measure.

- If the observed value of D^* is d_{obs}^* , then the p -value that describes this combined evidence is denoted by

$$p\text{-value}^* = \Pr(D^* \geq d_{obs}^* | H_0 \text{ is true})$$

- $p\text{-value}^*$ will necessarily be larger than $p\text{-value}_{min}$ because
 - $p\text{-value}_{min}$ is the smallest p -value among $p_1, p_2, \dots, p_K$ and so
 - $p\text{-value}_{min}$ exaggerates the evidence against the hypothesis and is misleading as a p -value.
- Given the data, all probabilities are proportions, hence D^* and $p\text{-value}^*$ can be calculated.

Estimating d_{obs}^*

- Suppose we have K discrepancy measures $D_1, D_2, \dots, D_K$.
 - The combined discrepancy measure is

Same M from previous notes

$$D^* = 1 - \left(\min_{k=1,2,\dots,K} p_k \right)$$

- For $i = 1, \dots, M_{inner}$ and each discrepancy $k = 1, \dots, K$
 - randomly mix the two sub-populations $\mathcal{P}_1$ and $\mathcal{P}_2$ yielding $\mathcal{P}_{1,i}^*$ and $\mathcal{P}_{2,i}^*$
 - calculate

$$d_{k,i} = D_k(\mathcal{P}_{1,i}^*, \mathcal{P}_{2,i}^*)$$

- then we estimate each p -value labelled as p_k with

$$p_k = \frac{1}{M_{inner}} \sum_{i=1}^{M_{inner}} \mathbb{I}\{d_{k,i} \geq d_{k,obs}\}$$

- Finally, we estimate $p\text{-value}_{min}$ and hence $d_{obs}^* \rightarrow 1 - p\text{-value}_{min}$

$$\downarrow$$

$$\min_{k=1,\dots,K} p_k$$

Estimating p -value*

- In order to estimate p -value* we need an idea of how extreme $\hat{d}_{obs}^*$ is if H_0 were true
- Thus we need to generate a distribution for D^* so that p -value* can be estimated.

* This is achieved by repeating the the steps above over and over again.

- Conceptually this requires nested looping; an *inner* loop to calculate $\hat{d}_{obs}^*$, and an *outer* loop to generate a distribution of $\hat{d}_{obs}^*$ values

- In particular, we repeat the following steps M_{outer} times:
 - randomly construct two sub-populations and
 - estimate d_j^* by the same procedure used to calculate d_{obs}^* (see above)

- then we estimate the p -value* with

$$\widehat{p\text{-value}^*} = \frac{1}{M_{outer}} \sum_{j=1}^{M_{outer}} I(d_j^* \geq \hat{d}_{obs}^*)$$

R Code & Example

- Below is a function that will calculate p -value*, the p -value that accounts for multiple discrepancy measures in the context of multiple testing.
- Notice that throughout the code the functions `sapply`, `Map`, and `Reduce` have been used instead of nested loops for efficiency.

```
calculateSLmulti <- function(pop, discrepancies, M_outer = 1000, M_inner){  
  #pop is a list whose two members are two sub-populations  
  if (missing(M_inner)) M_inner <- M_outer  
  ## Local function to calculate the significance levels  
  ## over the discrepancies and return their minimum  
  
  getSLmin <- function(basePop, discrepancies, M) {  
    observedVals <- sapply(discrepancies,  
                           FUN = function(discrepancy) {discrepancy(basePop)})  
  
    K <- length(discrepancies)
```

This calculates
 p -value*

```

total <- Reduce(function(counts, i){
  #mixRandomly mixes the two populations randomly, so the new sub-populations are indistinguishable
  NewPop <- mixRandomly(basePop)

  ## calculate the discrepancy and counts
  Map(function(k) {
    Dk <- discrepancies[[k]](NewPop)
    if (Dk >= observedVals[k]) counts[k] <- counts[k] + 1 },
    1:K)
  counts
},
1:M, init = numeric(length=K))

SLs <- total/M
min(SLs)
}

SLmin <- getSLmin(pop, discrepancies, M_inner)

total <- Reduce(function(count, m){
  basePop <- mixRandomly(pop)
  if (getSLmin(basePop, discrepancies, M_inner) <= SLmin) count + 1 else count
}, 1:M_outer, init = 0)

SLstar <- total/M_outer
SLstar
}

```

- As an example, let's compare the lengths of sharks in Australian versus American encounters.
 - We examine two discrepancies: one that compares averages and another that compares standard deviations.
 - Strictly speaking, this is a multiple testing scenario because we are testing the same null hypothesis with multiple (two) discrepancy measures.

```

getSDRatioFn <- function(variate) {
  function(pop) {sd(pop$pop1[, variate])/sd(pop$pop2[, variate])}
}

getAbsAveDiffsFn <- function(variate) {
  function(pop) {abs(mean(pop$pop1[, variate]) - mean(pop$pop2[, variate]))}
}

discrepancies <- list(getAbsAveDiffsFn("Length"), getSDRatioFn("Length"))

sharks <- read.csv("../Data/sharks.csv",
  header = TRUE)

pop <- list(pop1 = subset(sharks, sharks$Australia == 1),

```

```

pop2 = subset(sharks, sharks$USA == 1))

### The following takes a long time (about 20 minutes)
### for M_outer = M_inner = 1,000 say
### So for illustration much smaller values than would be sensible are
### used here
set.seed(341)
SLstar=calculateSLmulti(pop, discrepancies, M_outer = 100, M_inner=100)
SLstar

```

```
## [1] 0.56
```

*p-value**

- Since the p -value is large (0.56), there is no evidence against the hypothesis that the US and Australian encounters were randomly drawn from the same population based on the average and standard deviation of shark lengths.

Exercise: increase the M_{outer} and M_{inner} values above to get a more accurate estimate of the p -value, but note that this is computationally intensive.

Summary

- Here we combine several discrepancy measures into a single test of significance using the minimum p -value for each individual test.
- This procedure requires simulation to obtain d_{obs}^* and each d_j^* .

Exercises

- From the Inference Exercises try problem 8.